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ANNUAL REVIEWS

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The Enemies of Reliable and Useful Clinical Prediction Models: A Review of Statistical and Scientific Challenges

Ben Van Calster,[1,2] Maarten van Smeden,[3] Wouter van Amsterdam,[3] Maarten Coemans,[4] Laure Wynants,[1,2,5] and Ewout W. Steyerberg[3]

[1]Leuven Unit for Health Technology Assessment Research (LUHTAR), KU Leuven, Leuven, Belgium; email: ben.vancalster@kuleuven.be

[2]Department of Development and Regeneration, KU Leuven, Leuven, Belgium

[3]Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht, the Netherlands; email: m.vansmeden@umcutrecht.nl, w.a.c.vanamsterdam-3@umcutrecht.nl, e.w.steyerberg@umcutrecht.nl

[4]Department of Microbiology, Immunology and Transplantation, KU Leuven, Leuven, Belgium; email: maarten.coemans@kuleuven.be

[5]Department of Epidemiology, CAPHRI (Care and Public Health Research Institute), Maastricht University, Maastricht, the Netherlands; email: laure.wynants@maastrichtuniversity.nl

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clinical prediction models, methodology, model development, external validation, impact studies, model deployment

Abstract

The current status of applied clinical prediction modeling is poor. Many models are developed with suboptimal methods and are not evaluated, and hence have little impact on clinical care. We review 12 challenges—provocatively labeled enemies—that jeopardize the creation of prediction models that make it to clinical practice to improve treatment decisions and clinical outcomes for individual patients. The challenges cover four areas: context, data, design and analysis, and scientific culture. We provide negative examples and recommendations for improvement, but also highlight positive examples and developments. Greater awareness of the complexities surrounding clinical prediction modeling is needed among researchers, funding

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agencies, health professionals as end users, and all of us as potential patients. To improve the utility of prediction models for healthcare and society, we need fewer but better models as well as more resources for model validation, impact assessment, and implementation.

1. INTRODUCTION

Prediction modeling is an active area of medical research. Typically, the outcome of clinical prediction models relates to whether an event is present or not (diagnostic prediction) or to whether an event will develop within a given time horizon (prognostic prediction). Unfortunately, systematic reviews of prediction models consistently show us that we face an overload of untrustworthy models that will never make it to clinical practice. A few recent examples are provided by Damen et al. (2016), Whittle et al. (2018), Bellou et al. (2019), Christodoulou et al. (2019), Gerry et al. (2020), Park et al. (2020), Groot et al. (2021), Wynants et al. (2020), Dhiman et al. (2021), Andaur Navarro et al. (2021), Gao et al. (2023), Hudda et al. (2023), Barreñada et al. (2024), and Tsegaye et al. (2025). The sidebar titled The Poor State of Affairs for Prediction Models: COVID-19 illustrates this for prediction models in the context of COVID-19. Systematic reviews consistently report that only a small portion of developed models have been externally validated. When models are externally validated, substantially worse predictive performance (in terms of discrimination and/or calibration) is generally found than what was reported in the original model development paper (Wessler et al. 2021, Yu et al. 2022).

Discrimination: the ability to distinguish between individuals with the event and individuals without the event; preferably assessed by the concordance statistic (c-statistic)

Calibration: how well estimated risks correspond to observed proportions of individuals with the event; preferably assessed with flexible calibration plots

Of the even smaller group of models with satisfactory predictive performance in external validation studies, even fewer undergo an actual evaluation of clinical impact (Kappen et al. 2018, van de Sande et al. 2021). Only a tiny fraction of the prediction models that have been developed carefully reach the stage of deployment into clinical practice (**Figure 1**). On the other hand, several prediction models have been implemented in practice without sufficient evidence of clinical benefit (Steyerberg et al. 2013).

Clear recognition of the challenges of obtaining reliable and clinically useful prediction models will allow the prediction research community to take targeted action to improve the current state of affairs in clinical prediction modeling. In this review, we aim to present a broad overview of 12 key challenges—or enemies, to put it provocatively—that arise in the context of prediction modeling in medicine (**Table 1**). We define four broad areas of challenges for the trajectory from development to deployment of clinical prediction models, based on extensive literature review: context, data, study design and analysis, and scientific culture. For each challenge, we provide

THE POOR STATE OF AFFAIRS FOR PREDICTION MODELS: COVID-19

On March 24, 2020, 31 diagnostic and prognostic prediction models for COVID-19 had already been published, according to the first report of a living systematic review (Wynants et al. 2020). All models were at high risk of bias. On July 1, 2020, for the third update of the review, the number of developed prediction models had increased to 208, all at high or unclear risk of bias. On February 17, 2021, the living systematic review was updated for the fourth and last time, focusing on prognostic models only. Among 381 prognostic models, 374 were at high or unclear risk of bias and only 7 at low risk of bias. Another COVID-19-focused review included 62 diagnostic deep learning models based on chest radiographs and computed tomography images (Roberts et al. 2021). It concluded that "none of the models identified are of potential clinical use due to methodological flaws and/or underlying biases" (Roberts et al. 2021, p. 199).

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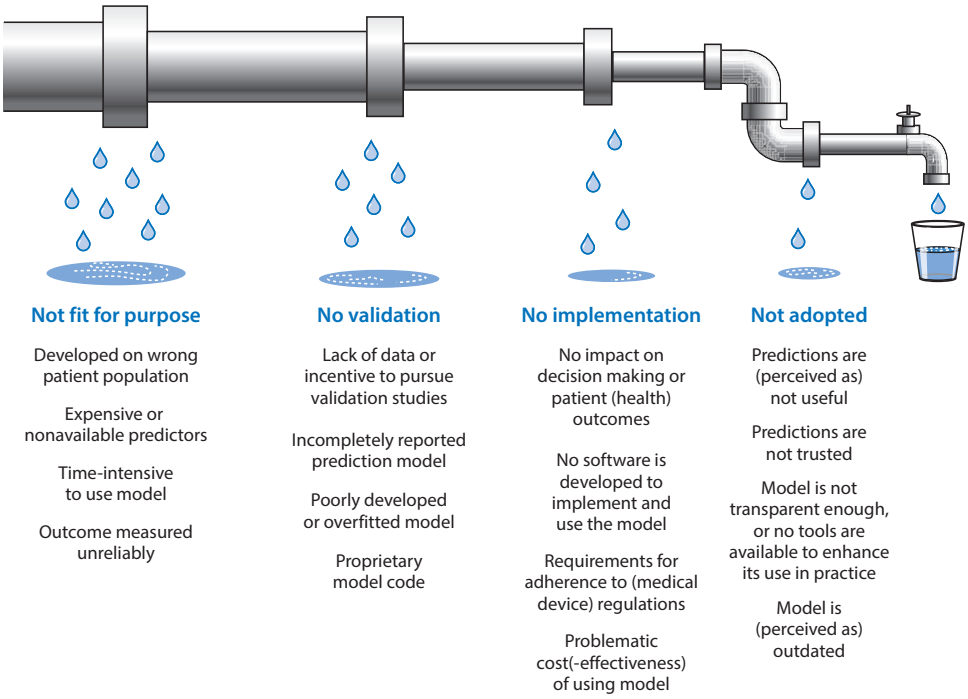

Figure 1

Visualization of the difficulties from development to successful deployment of prediction models. Reproduced from van Royen et al. (2022), with permission of the European Respiratory Society (ERS) (2025).

recommendations for improvement and positive examples or evolutions. Each enemy is illustrated for the COVID-19 prediction modeling field.

2. ENEMIES RELATED TO CONTEXT

The first set of enemies refers to the reality within which clinical prediction modeling operates. These issues cannot be avoided, but they need to be considered when conducting prediction modeling research.

Table 1 Twelve enemies for reliable and useful clinical prediction models

Domain	Enemy
Context	1. Low signal-to-noise ratios are common.
	2. Reliable individual risk estimation is impossible.
	3. Heterogeneity in time and place affects transportability.
Data	4. Measurement problems challenge model quality.
	5. Gold-standard diagnoses are elusive.
	6. Follow-up events distort prognosis.
Design and analysis	7. Poor methodology leads to useless models.
	8. Poor reporting limits usability of models.
	9. Model impact on clinical practice is poorly studied.
Scientific culture	10. Discovery is prioritized over confirmation.
	11. Education in data science is too narrow.
	12. Commercial interests challenge transparency and progress.

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2.1. Enemy 1: Low Signal-to-Noise Ratios Are Common

For a prediction model, the outcome Y (1 for event, 0 for nonevent) can be written as $p(Y|\mathbf{X}) + \varepsilon$, where $p(Y|\mathbf{X})$ is the true probability that $Y = 1$ conditional on a set of predictors $\mathbf{X}$ that is used in fitting the model. Signal then refers to explainable variation, the variability in Y that can be related to $\mathbf{X}$; this is the variance in $p(Y|\mathbf{X})$ over the distribution of $\mathbf{X}$, $\mathrm{Var}_{\mathbf{X}}[p(Y|\mathbf{X})]$. The term ε refers to the unexplainable variation, or noise, the remaining variability in Y conditional on $\mathbf{X}$, $E_{\mathbf{X}}[\mathrm{Var}(Y|\mathbf{X})]$. Noise can arise due to the inherent random nature of Y, or to the omission of unmeasured or unknown predictors with predictive value. In clinical medicine, the signal-to-noise ratio is often modest to low (Andrade 2013, Ennis et al. 1998, Makridakis et al. 2018): Predictors cannot easily separate individuals with the event from individuals without, given that many pathways can lead to a particular disease and diseases express in various ways.

The prediction modeler aims to identify the signal on an empirical dataset. This becomes harder with lower signal-to-noise ratios. The dataset also contains noise that is inversely related to sample size, through random sampling variation. Mistaking dataset noise for signal leads to overfitting of the data used to develop a model, which reduces model performance in new data from the same population. Balancing between signal and dataset noise is related to the bias–variance tradeoff. The mean squared error (MSE) is the average squared difference between y and $\hat{p}(Y|\mathbf{X}, D)$, the estimated probability that $Y = 1$ conditional on $\mathbf{X}$ across repetitions of the same model development study in different training datasets D of the same sample size. A classic result is that the MSE can be decomposed into squared bias, variance, and irreducible error (Friedman 1997):

$$E_D\left[Y - \hat{p}(Y|\mathbf{X}, D)\right]^2 = \left\{p(Y|\mathbf{X}) - E_D\left[\hat{p}(Y|\mathbf{X}, D)\right]\right\}^2 + E_D\left\{\hat{p}(Y|\mathbf{X}, D) - E_D\left[\hat{p}(Y|\mathbf{X}, D)\right]\right\}^2 + \mathrm{Var}(\varepsilon|\mathbf{X}).$$

The bias (the first term on the right-hand side of the equation) refers to the extent to which the model, on average, deviates from the true signal. Variance (the second term) refers to the extent to which fitted models vary as a function of D. Irreducible error (the third term) refers to unexplainable variation and hence to the signal-to-noise ratio.

Overfitting is the result of high model variance: The model is too dependent on D and tends to capture random dataset noise. This means that the model is too flexible for the training data. However, if the model is not flexible enough, we miss signal, resulting in bias (underfitting). Less flexible algorithms, such as generalized linear regression models, tend to have lower variance but higher bias compared with more flexible algorithms, such as random forests, gradient boosting methods, and neural networks. Tuning of hyperparameters is a strategy to balance over- and underfitting.

2.1.1. Recommendations: modest flexibility, exploiting domain knowledge, and larger sample sizes. A balance needs to be sought between the flexibility of a model development approach and the available quantity and quality of available data. Many have argued that, in low signal-to-noise settings, simple methods such as regression are preferable over more flexible alternatives that may be more prone to mistaking dataset noise for signal (Ennis et al. 1998, Hand 2006, Makridakis et al. 2018). A systematic review of studies developing prediction models using flexible machine learning and regression on tabular data concluded that flexible models do not have better performance on average (Christodoulou et al. 2019). More work is needed to compare algorithms on large datasets from electronic health records that typically include large sets of potential predictors (Rajkomar et al. 2019).

Furthermore, we recommend using available domain knowledge to prespecify predictors where reasonably possible. Strong reliance on data-driven variable selection methods tends to yield unstable selections that do not perform well in validation studies (Heinze et al. 2018, Steyerberg et al. 2018). Penalization, e.g., with the lasso, reduces overfitting but is unstable in small datasets (Van Calster et al. 2020b).

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The required sample size for developing new regression-based prediction models should be estimated in relation to the anticipated signal (e.g., via the R-squared) (Riley et al. 2020). More research is needed on the required sample size for machine learning algorithms (Rajput et al. 2023), as is a better understanding of the link between algorithm complexity and performance (Belkin et al. 2019, Curth et al. 2023). For any algorithm, adaptive sample size procedures can be considered (Christodoulou et al. 2021).

2.1.2. Positive examples/evolutions. Relevant effect sizes are sometimes found for simple-to-measure variables such as age. For example, age is strongly prognostic for mortality in surgery (Finlayson & Birkmeyer 2001). In that sense, domain knowledge can often help to delineate a simple core model with broad validity, such as for predicting poor outcomes after traumatic brain injury (Steyerberg et al. 2008). While some underfitting might occur by not including certain relevant predictors, core models help to avoid overfitting and increase generalizability and transportability.

There are several large collaborative initiatives that obtain large sample sizes, which further reduces overfitting. The QRisk initiative is one example (Hippisley-Cox et al. 2024).

2.2. Enemy 2: Reliable Individual Risk Estimation Is Impossible

Risk estimates, i.e., estimated probabilities of the event for an individual, are sought after for patient counseling and for making informed clinical decisions (Hunink et al. 2014). It is therefore essential that risk estimates are calibrated (Van Calster et al. 2019a). Although we ideally want to provide every individual with a correct risk estimate, individual risk is an elusive concept due to the reference class problem: Individuals belong to an infinite number of groups defined by different sets of predictors (Stern 2025). Rather, risk estimates from prediction models refer to the probability of the outcome event for the subgroup of individuals with the same values for the model's predictors (covariate pattern). Notably, this subgroup can quickly become very small when more predictors are added, especially for continuous predictors.

Obtaining correct subgroup probabilities for all possible covariate patterns, which has been labeled strong calibration, is unrealistic (Van Calster et al. 2016). Doing so would imply that the true model for the included predictors has been found, which is utopic (enemy 1). This is in line with recent research showing that risk estimates from prediction models are highly unstable even when considering a small and fixed set of predictors (Pate et al. 2019, Barreñada et al. 2025, Riley et al. 2025). The impossibility of reliably estimating subgroup probabilities also implies that many models can be made with more or less similar overall performance but with huge differences in individual predictions (Ledger et al. 2023, Li et al. 2020, Steyerberg et al. 2005).

2.2.1. Recommendations: moderate calibration and cautious interpretation. Practically, we should aim for models where the estimated risk is calibrated for every value of the estimated risk (moderate calibration): For example, among all individuals with an estimated risk of 20%, we would observe that 20 per 100 have or develop the event (Van Calster et al. 2016). This can be assessed by a flexible calibration plot (see **Figure 2** for an example). Even if moderately calibrated, the observed proportion (20/100) may not be correct for any individual receiving this risk estimate (20%) because different covariate patterns may lead to the same risk estimate. While this focus on moderate calibration may feel uncomfortable, it guarantees clinically useful decisions for the population (Van Calster et al. 2016).

The instability of risk estimates necessitates caution when presenting individual risk estimates for discussions with patients for counseling and decision-making.

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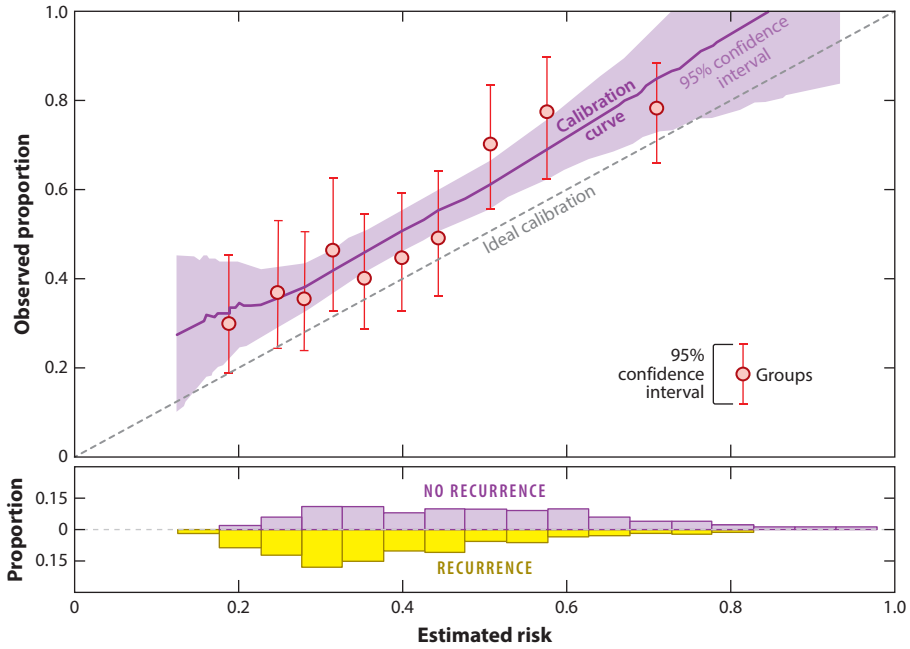

Figure 2

Example calibration plot for a model that estimates the risk of recurrence within five years after being diagnosed with primary breast cancer. The *x*-axis represents the estimated risk (including histograms for patients with and without recurrence), and the *y*-axis shows the observed proportion. The purple line is a flexible calibration curve linking estimated risks with observed proportions. This plot shows that observed proportions were higher than estimated risks, indicating that risks were underestimated. The red dots are a grouped version of the plot, showing the observed proportion of recurrence by deciles of estimated risk. Adapted from Riley et al. (2024a).

2.2.2. Positive examples/evolutions. Although further improvements are clearly needed, an increasing number of validation studies report calibration plots and/or summary statistics for calibration.

Increasing attention is being paid to uncertainty in risk estimates, for example, by efforts to quantify and visualize instability and uncertainty and by developing methods that indicate when a prediction is too uncertain (Kompa et al. 2021, Riley et al. 2025, Thomassen et al. 2024). Unfortunately, these methods largely target uncertainty related to sample size issues, not other less visible or quantifiable sources, such as model uncertainty (Hüllermeier & Waegeman 2021).

2.3. Enemy 3: Heterogeneity in Time and Place Affects Transportability

Clinical reality is heterogeneous (Van Calster et al. 2023). Countries, and even regions within the same country, often have marked differences in their healthcare systems (Agniel et al. 2018). Clinical sites (e.g., hospitals or general practices) also differ with respect to their patient populations, clinical protocols, availability of clinical expertise, and availability of medical equipment (Badgeley et al. 2019, Futoma et al. 2020). Heterogeneity between settings means that associations between predictors and an outcome can be context dependent. A prediction model may be accurate and useful for one medical setting but not for another. Patient populations are not static, but tend to evolve due to changes in lifestyle or healthcare protocols over time (Challen et al. 2019, Davis et al. 2020).

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These contextual dependencies have been recognized in the data science literature under the term "data drift" (Sahiner et al. 2023). Data drift affects a model's ability to make accurate predictions. Calibration performance usually deteriorates, with more over- or underestimation of estimated risks in different settings or at later time points. Discrimination often deteriorates as well, but it may also improve when a new population is more heterogeneous or contains a larger proportion of easy-to-predict individuals. Systematic reviews of diagnostic and prognostic models often report clear performance heterogeneity between settings (e.g., de Jong et al. 2022, Wessler et al. 2021).

Several journals prefer that papers in which new prediction models are presented include an external validation (Nevin 2018). While this is an intuitive requirement, a single external validation is insufficient to claim model validity. Furthermore, requiring a successful external validation study may elicit selective reporting.

2.3.1. Recommendations: more external validation and updating. Given the heterogeneous reality, researchers should clearly describe the specifics of the intended use of a newly developed model. Multiple high-quality external validations from a wide variety of settings are required to study and understand heterogeneity in predictive performance. The typical performance range for the c-statistic may be 0.1 or more (van Leeuwen et al. 2025). Validation studies further provide the basis for model updating (Binuya et al. 2022).

More attention should be given to developing and validating models on participants from multiple clusters to quantify heterogeneity rather than claim validity (Riley et al. 2016). Cluster (or setting) can refer to a site, hospital, center, practice, region, country, or combination of these. Methods to address heterogeneity in clustered data include internal-external (or leave-center-out) cross-validation in individual patient data meta-analyses, random effects modeling, or the use of cluster-level predictors (Takada et al. 2021; Wynants et al. 2018, 2019).

To address performance shifts, sites can consider continuous monitoring of performance and updating of models (Davis et al. 2020, Jenkins et al. 2021, Youssef et al. 2023). To this end, resources are required for high-quality data collection and for methodological expertise related to performance monitoring and model updating. Also, sites may not see enough patients to make this feasible. It is therefore important to study reasons for performance heterogeneity, which may allow more efficient and general updating efforts rather than the unrealistic goal of having local models at each site (Swaminathan et al. 2025).

Specific care should be taken when evaluating prediction models that are used to guide clinical decisions that change the probability of the outcome that is predicted (a prediction paradox; see enemy 6) (Lenert et al. 2019, van Amsterdam et al. 2023). For example, giving cholesterol-lowering medication to patients with a high estimated risk of cardiovascular disease may lead to reduced risks of a heart attack. In these cases, a decrease in performance may sometimes indicate a clinically successful deployment.

2.3.2. Positive examples/evolutions. Heterogeneity in regression coefficients and performance can be quantified in meta-analyses (Riley et al. 2016, Steyerberg et al. 2019). For example, a systematic review of external validation studies of the ADNEX (Assessment of Different NEoplasias in the adneXa) model to diagnose ovarian cancer quantified heterogeneity in the c-statistic (for binary outcomes, this equals the area under the receiver operating characteristic curve, or AUROC) and the net benefit statistic for utility (Barreñada et al. 2024). A meta-analysis of 9,202 tumors from 43 hospitals in 18 countries summarized in 21 studies reported an overall c-statistic of 0.93 and a 95% prediction interval of 0.85 to 0.98 as the estimated range within which the c-statistic lies in 95% of the hospitals. The overall net benefit of ADNEX at a threshold of 10%

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on the estimated risk of malignancy was 0.28, with a 95% prediction interval of 0.05 to 0.65 and an estimated probability of 0.95 that ADNEX has clinical utility in a new hospital.

Another example is that Memorial Sloan Kettering Cancer Center regularly updates its prediction models. By including the most recent patient data, they aim to make prediction models "more like Netflix" (Vickers et al. 2017).

3. ENEMIES RELATED TO DATA

The second set of enemies are related to data issues in prediction modeling studies. Although these challenges cannot simply be solved, careful focus on the magnitude of the problem and best possible handling strategies may improve results. In particular, the quality and applicability of developed models and the risk of bias of estimated model performance can improve.

3.1. Enemy 4: Measurement Problems Challenge Model Quality

In this era of flexible algorithms and large datasets, data and measurement quality remain important (Varoquaux & Cheplygina 2022, Willemink et al. 2020). Incomplete information on predictor variables often is not discussed or is handled inappropriately (see also the systematic reviews cited above), which may lead to biased models or badly estimated risks. Missing data are a particularly common problem for routinely collected data from electronic health records. Missing data rarely occur completely at random. For example, patients with worse health status may have fewer (e.g., in electronic health records) or more (e.g., in longitudinal follow-up data) missing values compared with patients with better health (Rusanov et al. 2014, Vanneste et al. 2016). Using a default analysis strategy, such as complete case analysis, may then lead to an underrepresentation of complex and/or sicker patients. In most cases, approaches for handling missing data, such as statistical imputation methods, help to exploit all information at model development and lead to better models (Sisk et al. 2023).

The measurement of a predictor impacts model development and validation. It is clear that random measurement error negatively affects discrimination performance. However, predictor measurement heterogeneity, i.e., systematic differences in measurement procedures between data for model development and external validation, is of particular importance (Luijken et al. 2019). Predictor measurement heterogeneity tends to induce miscalibration and change discrimination, sometimes leading to models without clinical utility in certain sites.

3.1.1. Recommendations: define measurements and context-specific missing data handling. Procedures for measuring predictors should be carefully described, such that clear comparisons between development and validation setting can be made (Cox et al. 2025). If standardized definitions are available, these should be used where possible to reduce predictor measurement heterogeneity. For appropriate missing data handling, it is important to consider the missingness mechanism and the specifics of the targeted future deployment of the prediction model (how, where, for whom, and the extent to which missing values will be present or can be avoided). For example, real-time deployment of prediction models, such as early warning scores in hospitalized patients using electronic health records, should deal with missing values in real time as well. These considerations help to determine which approach for missing data might be defendable (Tsvetanova et al. 2024). Approaches include (multiple) imputation, submodeling, marginalization, missing indicator methods, complete case analysis, and others (Hoogland et al. 2020, Sisk et al. 2023).

3.1.2. Positive examples/evolutions. The International Ovarian Tumor Analysis (IOTA) group published a terms and definitions statement to explain how ultrasound measurements

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of an ovarian tumor can be defined before embarking on prediction model development (Timmerman et al. 2000). This statement has become a standard in the field, reducing measurement heterogeneity between sites.

Interobserver agreement analysis of potential predictors has been undertaken prior to developing a prediction model for fracture in patients with acute knee injuries (Stiell et al. 1995). A multicenter study to develop a risk model to diagnose ovarian cancer, where an agreement analysis was not possible, used the residual intraclass correlation to more broadly investigate inter-center variations in predictor measurements (Wynants et al. 2013).

Because most research on missing data handling focused on causal research questions, research in the context of prediction modeling has increased in recent years, leading to targeted recommendations such as the consideration of whether missing values are allowed at deployment (e.g., when making real-time predictions based on electronic health records data) or not (Sisk et al. 2023).

When validating models, results can be shown using different missing data handling approaches, such as complete case analysis versus multiple imputation for a model predicting survival in patients with HIV/AIDS (Wang et al. 2020).

3.2. Enemy 5: Gold-Standard Diagnoses Are Elusive

Diagnostic prediction models require the definition of presence of disease. Ideally, an error-free gold standard defines the diagnosis. A more practical concept is to define a clinical reference standard, the best available approach to define the diagnostic outcome. The reference standard may be invasive and/or costly to perform and hence selectively performed (Naaktgeboren et al. 2016). For example, if surgery is required to confirm a diagnosis, it may be impossible to impose surgery for patients where this would otherwise not be done. Researchers may then only include patients for whom the reference standard was available as per clinical indication, which leads to selection bias due to partial verification. The included sample may contain a higher proportion of suspicious cases compared with the target population. Such selection jeopardizes the relevance of diagnostic models for the intended target population. Alternatively, if patients without the invasive reference standard are included as well, the outcome has to be defined using a different—and often less accurate—approach, leading to differential verification (de Groot et al. 2011).

In some diagnostic models, the outcome is based on (fairly) objective reference standards based on procedures such as histological examination of tissue biopsies or death certificates. Other models predict outcomes that are more subjective, even apart from patient-reported outcomes (Weinfurt & Reeve 2022). An example is described in the sidebar titled Illustration of Problems with Diagnostic Labels in the Context of Kidney Transplant Rejection. A particular case involves prediction modeling based on medical images (Chen et al. 2021). Here, the reference standard is often based on (consensus) clinical judgment and/or manual or automatic classification of free text reports from the managing clinician. The quality of such judgments can vary substantially, and interobserver agreement can be limited. This may lead to poor and unstable outcome labels.

3.2.1. Recommendations: careful choice of reference standard and analysis method. An objective reference standard that is clinically meaningful should be used where possible. For medical images, use structured reporting and assess inter-rater variability. If a reference standard cannot be obtained in all participants, avoid excluding the cases where it cannot if they are relevant for the prediction estimand (what you want to predict when, in whom, and for what purpose). Instead, consider using a differential reference standard, with a less invasive/costly method to obtain the outcome status for a subset of patients. In that case, methods for handling differential verification should be applied (de Groot et al. 2011, Naaktgeboren et al. 2016).

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ILLUSTRATION OF PROBLEMS WITH DIAGNOSTIC LABELS IN THE CONTEXT OF KIDNEY TRANSPLANT REJECTION

Kidney rejection after transplantation is diagnosed using histopathological assessment of allograft biopsies, which are often performed as per clinical indication. This assessment is not straightforward, because several rejection phenotypes exist that can overlap. Every two years, the Banff classification rules for rejection are discussed based on advancing insights and updated if necessary. Although the Banff rules serve as the current reference standard, it has flaws due to its dependence on pathologist expertise, interobserver variability, and artificial dichotomization of a complex disease spectrum.

To overcome this, the Banff Digital Pathology Working Group started to study the use of deep learning to detect rejection more objectively: "It is possible that AI/ML [artificial intelligence/machine learning] will serve as a 'gold standard' in some sense, although we foresee AI/ML augmenting pathologists rather than replacing them as the 'gold standard'" (Farris et al. 2023, p. 7). A first study reported a c-statistic of 0.76 for predicting rejection (Kers et al. 2022). Another considered approach is to add biopsy-based molecular transcripts analysis to the histopathological assessment of biopsies. While transcript analysis might be a more reproducible approach, limited progress has been made as yet (Naesens et al. 2024).

3.2.2. Positive examples/evolutions. The ADNEX model to diagnose ovarian cancer was developed on data from patients who were operated on such that the tumor outcome could be confirmed through histopathology (Van Calster et al. 2014). However, given that ADNEX should also be used to determine who needs surgery, a study (IOTA5) was set up that included all patients with regular follow-up visits, for patients without surgery. ADNEX has been validated on IOTA5 and is currently being updated such that the model is based on the correct target population (Van Calster et al. 2020a). Without surgery, follow-up information was used to classify the tumor as benign or malignant, and differential information was handled using multiple imputation and sensitivity analyses.

3.3. Enemy 6: Follow-Up Events Distort Prognosis

Prognostic modeling studies require follow-up information over a clinically relevant period of time. During follow-up, patients may experience a different, competing event that prohibits the occurrence of the event of interest. This poses distinct methodological challenges (Austin et al. 2021) but is sometimes ignored in prediction modeling studies (see, e.g., Gao et al. 2023, Wynants et al. 2020).

More importantly, many prognostic models do not properly account for treatments that are started, changed, or stopped during follow-up (van Geloven et al. 2020). The way such treatment events are handled affects the estimand or "predictimand," i.e., what the estimated risks actually estimate. Treatment changes are not random variables that simply happen, but are human-induced and often relate to the risk of the outcome that is predicted. Some prognostic models may intend to improve patient outcomes by guiding postprediction patient management. If such models change the healthcare process, the treatment paradox arises: Patients with a high estimated risk of the outcome event may undergo an intervention that reduces the risk (van Amsterdam et al. 2023). This paradox poses challenges when the implemented model needs to be evaluated and updated over time.

3.3.1. Recommendations: define estimands and consider a dynamic perspective. For prognostic outcomes, it is of particular importance to carefully define the estimand from time zero, the

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time at which the follow-up starts and the prediction is needed (Steyerberg et al. 2013). The predictimand framework can support appropriate methodological decisions for analyzing treatment initiation and stopping (Sperrin et al. 2018, van Geloven et al. 2020). This framework clarifies whether we consider predictions for patients under care as usual or a counterfactual situation, such as adhering to a typical treatment regimen, or leaving them untreated. Estimands also relate to competing events, which are present for most prognostic outcomes. Competing events need proper handling via the Fine–Gray model or, preferably, the combination of cause-specific hazard models (Austin et al. 2022).

Furthermore, it should be considered whether dynamic predictions (that are updated over time) are needed, rather than static prediction at time zero only (Putter & van Houwelingen 2022).

3.3.2. Positive examples/evolutions. Growing attention is being paid to estimands in prognosis, and decision-making on interventions guided by prediction models is being taken more seriously (Luijken et al. 2024, van Geloven et al. 2020). Guidance on validation prediction models with competing risks is available (van Geloven et al. 2022). Competing risks are being carefully handled in some prediction models (Albu et al. 2025, Coemans et al. 2025).

Dynamic prediction models have become more common in, for example, the prediction of wound healing at subsequent home visits (Thomassen et al. 2025), bloodstream infections in hospitalized patients (Albu et al. 2025), and symptom improvement at subsequent sessions of psychotherapy (Bone et al. 2021). Guidance on predictimands for dynamic prediction is available (Luijken et al. 2024).

4. ENEMIES RELATED TO DESIGN AND ANALYSIS

Designing and analyzing a prediction modeling study leads to specific challenges. Several design-related and analytic choices relate to context and data enemies, while others relate more specifically to computational choices or to how the study is reported to others.

4.1. Enemy 7: Poor Methodology Leads to Useless Models

Systematic reviews (see the Introduction) have consistently highlighted that the majority of prediction model development studies suffer from severe methodological flaws. With methodology, we here refer broadly to the methods used to set up, conduct, and analyze prediction model development or validation studies. Common flaws include nonrepresentative samples, small sample sizes, and poor analytic methods. An undervalued issue is blinding: Where possible, outcomes should be determined without knowledge of the predictors and vice versa. Some further examples of poor methodology that may cause overfitted, uninterpretable, or unusable models are listed in the sidebar titled Poor Performance with Common Statistical Modeling Approaches and in the examples from the COVID-19 prediction modeling field below. When methodology is poor, any claimed performance may be too optimistic, and risk estimates may be systematically over- or underestimated (Helmrich et al. 2022, Venema et al. 2021). Unnecessary dichotomization of continuous predictors is very common. While this may not always lead to overfitting, the loss of information reduces discriminative ability (leading to lower c-statistics).

Problematic model validation practices may yield optimistic impressions of performance. Examples include the absence of any kind of internal or external model validation, incorrect use of validation procedures (e.g., not freezing the developed model before applying it to the test set, not repeating all development procedures when using bootstrapping or cross-validation), and not assessing calibration performance at external validation (Christodoulou et al. 2019, Collins et al. 2024a).

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POOR PERFORMANCE WITH COMMON STATISTICAL MODELING APPROACHES

A leading medical journal (the *New England Journal of Medicine*) published a prediction model from the MMRPredict study in 2006 (Barnetson et al. 2006). The model was based on logistic regression analysis in a cohort with only 38 events among 870 participants. Stepwise selection was performed for model building, and predictors were dichotomized. A simulation study of this strategy showed instability in the selection of predictors, overfitting, and overoptimism with poor performance at validation (Steyerberg et al. 2018). Poor performance was confirmed in a large international validation study (Kastrinos et al. 2016). This case study illustrates that overfitting is common with small datasets used for model development. Standard techniques such as stepwise selection for regression models are a form of data torturing. More appropriate prediction selection procedures exist—for example, methods that leverage the lasso method, a method that performs selection with more appropriate penalization (Steyerberg et al. 2000, Tibshirani 2018). In addition, using domain knowledge and applying statistical thinking create more sensible predictor sets.

4.1.1. Recommendations: follow methodological guidance and collaborate with experts. Model development should follow existing guidance in the methodologic literature (see, e.g., Moons et al. 2015, Steyerberg 2019). Collaboration between statisticians with knowledge about prediction modeling and domain experts may often be needed to develop a promising and reliable prediction model.

It is important to properly design a model development study in terms of potential predictors and sample size in order to limit overfitting. Guidance for sample size determination for model development studies using regression modeling is available (Riley et al. 2020). Designing external validation studies also needs attention from researchers, with inclusion of calibration, to obtain trustworthy and meaningful results (Riley et al. 2024a,b). Research collaborations between sites for development and validation are useful to address heterogeneity (enemy 3), but also to increase sample size.

4.1.2. Positive examples/evolutions. Among the hundreds of models that were critically appraised during the living systematic review of models for COVID-19, a few models were identified that were developed well: QCovid, Carr, PRIEST, and ISARIC4C (Carr et al. 2021, Clift et al. 2020, Goodacre et al. 2021, Gupta et al. 2021). Not surprisingly, all models were developed on large clustered cohorts of individuals.

PROBAST is a structured tool that helps to assess the risk of bias and applicability of prediction models. Recently, the updated PROBAST+AI tool was released to specifically include models based on modern artificial intelligence methods (Moons et al. 2025).

The TRIPOD+AI (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis plus Artificial Intelligence) reporting guideline and PROBAST+AI tool have supporting elaboration and explanation documents that contain a treasure trove of information that helps to improve the methodological quality of prediction modeling studies (Debray et al. 2023, Moons et al. 2015, 2019). These documents are currently being updated with the latest knowledge.

The STRATOS (STRengthening Analytical Thinking for Observational Studies) initiative connects international experts to provide accessible guidance documents for designing and analyzing observational studies (Sauerbrei et al. 2014). STRATOS guidance targets applied data analysts with varying levels of statistical education, experience, and interests, as well as clinical researchers.

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4.2. Enemy 8: Poor Reporting Limits Usability of Models

Various systematic reviews (see those cited in the Introduction) show that reporting of prediction modeling papers is often incomplete and ambiguous. When poorly reported, readers and potential users cannot properly understand the why, when, how, and for whom of a newly developed model. This jeopardizes validation and implementation efforts. Further steps in the research pipeline, such as external validation and updating of models, require full reporting as well.

Full reporting relates to more general open science principles. Improvement is needed here, too, such as sharing analysis plans, analysis code, data, and the model itself to allow independent validation (enemies 9 and 10) (Collins et al. 2024c). For regression models, the model equation should be given, including the baseline risk (such as the intercept for logistic models or baseline hazard for survival models) (Van Calster et al. 2019b). For more complex machine learning algorithms, other solutions are needed, such as providing software objects. For any models, implementations such as web calculators can be provided, which require sufficient background information, disclaimers, and proper consideration of applicable legislation for medical devices (Kotlarz et al. 2025).

4.2.1. Recommendations: follow reporting guidelines. The TRIPOD+AI guideline provides a key tool to accompany the preparation of a scientific report on development or validation of a prediction model (Collins et al. 2024b). Specialized TRIPOD guidelines are available for clustered data and large language models (Debray et al. 2023, Gallifant et al. 2025; see also the EQUATOR website, **https://www.equator-network.org**). Such guidelines should be followed when starting to draft a paper, making them a tool to help with structuring the text rather than an extra chore at the end.

4.2.2. Positive examples/evolutions. The TRIPOD guidelines are included in an increasing number of editorial guidelines for journals (Heus et al. 2024). Recently, the original TRIPOD 2015 guidelines have been superseded by TRIPOD+AI, in an update with recent insights that includes items related to modern algorithms (Collins et al. 2024b). For example, considerations around fairness of prediction models have been added.

4.3. Enemy 9: Model Impact on Clinical Practice Is Poorly Studied

In a healthy life cycle of a clinical prediction model, evidence on clinical impact and implementation aspects is essential (de Hond et al. 2022). Currently, however, downstream consequences such as impact on care processes, efficiency gains, and improved patient outcome are speculative for most prediction models (Steyerberg et al. 2013, van Leeuwen et al. 2021, van Royen et al. 2022). Presumably, this is partially because impact and implementation studies are complex, require resources and time, often have negative results, and are considered as incremental rather than innovative research (enemy 10).

Two central points of discussion are the endpoints of impact and implementation studies, and the integration of the model into clinical care. While short-term process-related outcomes (e.g., unnecessary interventions avoided) are certainly relevant, long-term patient outcomes (e.g., mortality) are important but may suffer from loss to follow-up and smaller effect sizes (Kappen et al. 2018). Long-term outcomes are less often addressed in model impact studies; some results are described in the sidebar titled Findings of Two Systematic Reviews on Impact Studies. Results of impact studies depend strongly on the extent to which the model is integrated in care, such as the user-friendliness of the software integration, how model predictions are presented, and whether actionable recommendations are provided. These issues affect user adherence to model output.

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FINDINGS OF TWO SYSTEMATIC REVIEWS ON IMPACT STUDIES

A 2012 systematic review of impact studies based on (cluster) randomized trials identified 148 studies, of which 46 (31%) were multicenter studies (Bright et al. 2012). The following outcomes were investigated: healthcare process measures (i.e., patient management decisions regarding treatment or additional tests) in 128 (86%) studies, user acceptance/satisfaction in 38 (26%) studies, clinical outcomes (e.g., mortality, disease incidence, adverse events) in 29 (20%) studies, economic outcomes (costs or cost-effectiveness) in 27 (18%) studies, and user workload/efficiency in 11 (7%) studies.

A 2021 study evaluated 100 commercially available tools based on machine learning or deep learning to assist radiologists in interpreting medical images in clinical practice (van Leeuwen et al. 2021). Images were based on computed tomography, magnetic resonance, X-ray, mammography, and/or ultrasound. For 64 tools, no peer-reviewed evidence of efficacy or impact was found. For the remaining 36 tools, most of the studies were on diagnostic accuracy and clinical/technical feasibility. Impact on healthcare process measures was assessed for 7 tools, on patient outcomes for 1 tool, and economic outcomes for 2 tools.

4.3.1. Recommendations: impact and implementation studies are needed. We recommend that researchers consider various methods and designs that are available for impact assessment (Janse et al. 2024, van Royen et al. 2022). Decision curve analysis (DCA) is a tool that is readily applicable to validation data and that provides an early straightforward assessment of potential clinical utility (Van Calster et al. 2018, Vickers & Elkin 2006). DCA is based on the net benefit statistic for various plausible risk thresholds that may be used to select patients for intervention (**Figure 3**). Models that show promise in DCA can undergo further evaluation in empirical impact

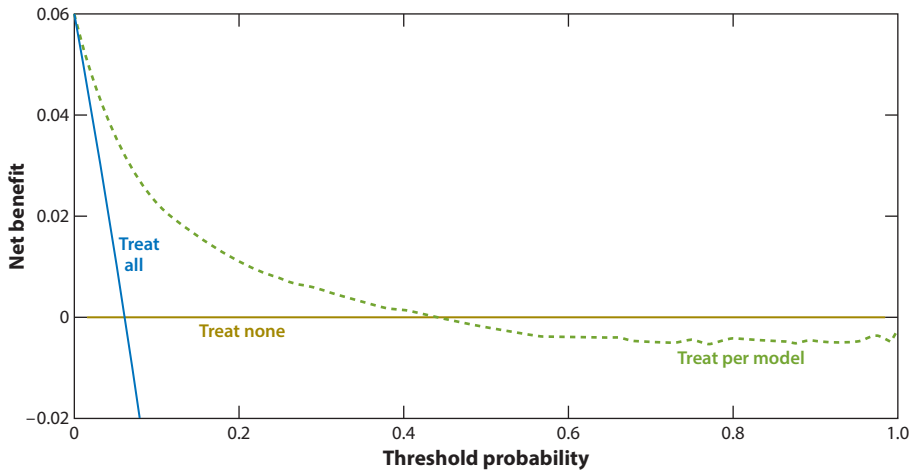

Figure 3

Example decision curve for a model that estimates the risk of 30-day mortality after acute myocardial infarction. The *x*-axis represents the threshold on the estimated risk that is used to suggest an intervention, and the *y*-axis is the net benefit of making the decisions using the misclassification costs that a threshold implies. The *x*-axis should ideally be limited to a range of reasonable thresholds for the problem at hand. The plot shows the net benefit of the model at various thresholds (*green*), of intervening in everyone at various misclassification costs (*blue*), and of intervening in no one (*gold*). A model has clinical utility if its net benefit is higher than that of treating all and treating none. Competing models can also be compared using net benefit. Adapted from Riley et al. (2024a).

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studies, which can provide invaluable information on user acceptance and human–model interaction. Standard, or preferably cluster randomized, trials are ideal to assess the impact of prediction models on patient outcomes, but they are tedious to conduct and demanding in terms of sample size. Before–after studies have also been used, and mixed methods studies have been used to obtain the views of clinicians who would use the model. The results of such studies can be integrated with information on the model's predictive performance, treatment effectiveness, and cost-effectiveness.

Deployment should be carefully prepared to ensure integration into the workflow of healthcare professionals (Brown et al. 2016, Kawamoto et al. 2005). Software implementation may require efforts in the realm of information technology, specifically for machine learning algorithms and for implementation of models into electronic health record systems. In addition, we recommend educating clinical end users on how they can use the model in the local context of their daily activities. Influencing their attitudes regarding data-driven care innovations typically requires strategic dissemination efforts (Meyers et al. 2012). Regulatory frameworks, such as the European Union Medical Device Regulation (MDR), and the European Union AI Act impose additional requirements when bringing a model into clinical practice, including setting up a quality management system to guarantee ongoing monitoring and maintenance of the model and software tool. Finally, medicolegal aspects need proper discussion (Cestonaro et al. 2023, McCradden & Stedman 2024).

4.3.2. Positive examples/evolutions. The following are examples of impact studies of clinical prediction models using various designs:

- DCA and cost-effectiveness analysis of models for ovarian cancer diagnosis (Westwood et al. 2018, Wynants et al. 2017)
- A cluster randomized trial of a deep learning model to screen for low ejection fraction on electrocardiograms (Yao et al. 2020)
- A randomized controlled trial of a model for serious bacterial infections in children with fever (de Vos-Kerkhof et al. 2015)
- A prospective observational paired-reader study of a deep learning algorithm to screen mammograms for abnormality (Dembrower et al. 2023)
- A decision-analytic analysis of models for pre-eclampsia in the first trimester of pregnancy (Strijbos et al. 2023)
- A before–after study with cost-effectiveness analysis of a model for first-trimester pregnancy complications (van Montfort et al. 2020)

5. ENEMIES RELATED TO SCIENTIFIC CULTURE

These enemies relate to the broader scientific culture in which researchers conduct their work. Individual researchers have limited capacity to tackle these challenges.

5.1. Enemy 10: Discovery Is Prioritized Over Confirmation

The current scientific culture is strongly focused on novelty at the expense of replication, confirmation, or fine-tuning. Research that is later found to be untrue may generate high citation rates, in contrast to later studies that invalidate or mitigate the claimed discoveries (Ioannidis 2005). Furthermore, academia consistently undervalues the methodological quality of research and values quantity, reputation, and financial income instead (Van Calster et al. 2021).

This culture pushes researchers toward low-quality studies in which new predictors are introduced in new prediction models, or quickly and poorly done validation studies. Prediction

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modeling studies are a popular outlet for academic researchers, as they appear relatively easy to perform and publish: The minimum required ingredients are a dataset and someone with basic knowledge of a statistical software program (Vickers & Cronin 2010). Problematic aspects in prediction model studies often pass the editorial process relatively easily, potentially due to the lack of knowledge about prediction modeling methods (enemy 11). Poor validation studies may be uninformative or even misleading by being unduly negative about a good model, or unduly positive about a poor model. This situation leads to publications that are primarily beneficial for the careers of the investigators, rather than for patients (whose tax money often funded the research). As noted above, systematic reviews of prediction models in various medical disciplines consistently observe an abundance of new models and a scarcity of validation studies in the literature, of which most are of poor quality.

5.1.1. Positive examples/evolutions. There are collaborative efforts to (*a*) develop models on large clustered datasets to avoid multiple smaller model development studies and (*b*) validate models on a large scale, for example, using individual participant meta-analysis (Allotey et al. 2024, Gulati et al. 2022).

Some prediction models, such as the ADNEX model for ovarian cancer diagnosis, are externally validated in multiple studies at various different locations (Barreñada et al. 2024).

5.1.2. What should change? There have been multiple calls to improve the quality of research in order to decrease the number of studies of little relevance and reduce financial losses incurred by poor research (Chalmers & Glasziou 2009, Ioannidis et al. 2014). For prediction modeling research, this would imply the development of fewer but better models, in combination with a strong focus on validation, impact, and implementation studies. Improving the research culture is a difficult, complex process involving multiple strongly entwined stakeholders. Researchers have an individual responsibility to follow good research practices. Academic institutions and funding bodies should help to increase the focus on clinical research methodology and model validation, updating, and impact studies. Journal editors should keep improving the peer review process for prediction modeling studies, requiring adherence to TRIPOD+AI or related guidelines.

5.2. Enemy 11: Education in Data Science Is Too Narrow

Statistical education for applied researchers, such as biostatisticians, data scientists, clinical epidemiologists, and clinical researchers, has traditionally focused on technical and computational aspects of data analysis (e.g., choosing analysis techniques based on whether variables are measured on a categorical or continuous scale). How data analysis should be embedded in, and should be dictated by, the broader real-life context of a specific study often receives little attention (Hicks & Irizarry 2018, Steel et al. 2019).

Education about regression methods is too often taught in the general context of statistical inference, implicitly assuming that the focus is on testing the (causal) effect of a particular variable on an outcome. Likewise, machine learning methods are often taught with a strong focus on computation at the expense of study design, data quality, and type of research question addressed.

5.2.1. Positive examples/evolutions. There is increasing agreement that clinical and statistical curricula need better training in methodology and statistical thinking (Aerts et al. 2021, Pierre et al. 2022, Woltenberg 2021). Some data science programs take broader perspectives beyond computing and technical issues (Alvarado & Zimmer 2025).

Prediction modeling and its methodological issues receives attention in some general medical journals, such as *BMJ*, and in targeted journals such as *Lancet Digital Health*, *NEJM-AI*, *npj Digital Medicine*, and the *Journal of the American Medical Informatics Association*.

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5.2.2. What should change? Educational programs should consider the broader clinical picture, including for prediction modeling research, before jumping to technicalities. The distinction between predictive and causal research questions should be embedded in curricula (Shmueli 2010). We argue that clinical prediction modeling is a field in its own right and should be taught as such, as done in targeted textbooks (Gerds & Kattan 2021, Riley et al. 2019, Steyerberg 2019). We recommend that education on clinical prediction modeling include sufficient attention for specification of the exact research question and estimand, study design and sample size, analysis choices, and inclusion of domain knowledge where possible. Ideally, this would be done using problem-based learning with realistic datasets.

5.3. Enemy 12: Commercial Interests Challenge Transparency and Progress

There is growing commercial interest in prediction models for healthcare (see, e.g., van Leeuwen et al. 2021). Academic spin-off companies, or large health technology companies, may provide prediction models through commercial software. While this allows financial investments in predictive analytics and may facilitate obtaining regulatory approval (Benjamens et al. 2020), it also poses a potential conflict of interest that may hamper transparency. External validation efforts are jeopardized by lack of transparency or even fee-per-service costs, resulting in (*a*) lack of knowledge about model performance and heterogeneity between settings, (*b*) the risk that real-world performance is worse than advertised, and (*c*) models that cannot be improved or updated (Mitchell et al. 2025, Van Calster et al. 2019b). For example, the proprietary sepsis model from Epic performed worse than advertised in an independent validation study (Wong et al. 2021).

5.3.1. Positive examples/evolutions. A combination of openness and commercial use needs to be sought, e.g., by public–private partnerships such as Evidencio, which hosts thousands of public medical algorithms with a free and fair use policy and encourages validation and updating.

For commercial deep learning models, modern validation infrastructures are being developed to facilitate external validation efforts for proprietary models (Ramwala et al. 2024).

Consortia such as the Trustworthy and Responsible AI Network (TRAIN) and TRAIN-Europe will help to push clinical prediction models, including commercial ones, toward being more qualitative, responsible, safe, transparent, and effective (van Genderen et al. 2025).

5.3.2. What should change? We need better procedures to combine commercial involvement and maximal transparency with respect to performing external validations, as well as reporting, to enable the continuous improvement of prediction models. The AI Act of the European Union recommends performance monitoring, external testing, and transparency in informing about capabilities and limitations of AI systems. Regulatory bodies should keep working to align their requirements accordingly.

6. THE ENEMIES APPLIED TO THE COVID-19 PREDICTION MODELING FIELD

We base this section mainly on the results of the now-finished living systematic review of diagnostic and prognostic prediction models for COVID-19 from Wynants et al. (2020), unless otherwise specified.

6.1. Low Signal-to-Noise Ratios Are Common

Prognostic COVID-19 studies typically had low sample size: The median sample size was 414 (interquartile range 172–1,505), and the median number of outcome events was 74 (36–207). 67% of studies did not meet required sample size based on events-per-variable criteria. 250 of 381 (66%)

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newly developed prognostic models lacked a proper assessment of overfitting. C-statistics of 0.77 to 0.93 for low risk of bias models for different populations and outcomes illustrate the modest signal-to-noise ratio.

6.2. Reliable Individual Risk Estimation Is Impossible

The quality of the risk estimates was not assessed in many reports, or assessed with uninformative procedures such as the Hosmer–Lemeshow test. Only 26% of prognostic models were evaluated in a calibration plot.

6.3. Heterogeneity in Time and Place Affects Transportability

The review identified 381 newly developed prognostic models for COVID-19, compared with 225 model validations. Many of the validations performed were not validations of the newly developed COVID-19 models, but validations of general-purpose prognostic scores in COVID-19 patients (e.g., NEWS2, qSOFA). Most were single-center external validations. Heterogeneity was only assessed in a few subsequent large collaborative efforts (Damen et al. 2024, de Jong et al. 2022).

6.4. Measurement Problems Challenge Model Quality

Missing values were ignored or unclearly reported. 34% of studies did not mention how missing data were handled, whereas 48% used inappropriate methods such as complete case analysis. Important complications such as pneumonia were measured differently across studies. Diagnoses of COVID-19 infections were done with different tests and work-ups. Predictors were often, and outcomes were sometimes, poorly defined.

6.5. Gold-Standard Diagnoses Are Elusive

Many diagnostic COVID-19 models were published early in the pandemic. The reference standard, the presence of COVID-19, was not unequivocally determined across studies. Studies often defined COVID-19 based on a positive polymerase chain reaction test, which was selectively used in critical periods; unclear definitions of pneumonia; radiological reports; and/or proxy outcomes, such as admission with severe respiratory disease (Griffith et al. 2020, Roberts et al. 2021). Sometimes, predictors (e.g., fever) were part of the outcome definition.

6.6. Follow-Up Events Distort Prognosis

A majority of studies (65%) did not specify the prediction horizon for the model. Other studies ignored follow-up time and modeled the outcome in a binary way, thereby ignoring the differences in available follow-up between patients. Some studies excluded patients who did not have the outcome event within the targeted prediction horizon. A majority of studies did not handle post-baseline treatments in way that is consistent with the intended model use (Prosepe et al. 2022).

6.7. Poor Methodology Leads to Useless Models

Most models were considered at high risk of being biased due to methodological and analytical shortcomings. Problems included selection bias, small sample size in combination with complex models, ignoring domain knowledge when selecting predictors, failure to properly evaluate model performance, lack of proper discussion and handling of missing data, using predictors that were not available at prediction time in practice, and failing to evaluate the calibration performance. Information was often lost by unnecessarily dichotomizing continuous predictor variables. Some studies

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used problematic designs in which cases and controls were sampled from different populations (e.g., adults versus children, or Chinese versus American centers).

6.8. Poor Reporting Limits Usability of Models

Reporting was unclear and incomplete for many published models, including basic aspects related to sample size, number of outcome events, variable selection methods, or missing data handling (Hudda et al. 2023). A majority of studies did not make models available for use by providing a formula, calculator, or software object.

6.9. Model Impact on Clinical Practice is Poorly Studied

Time was insufficient to assess any of the COVID models empirically for their positive or negative impact on care. Nonetheless, there is evidence that some models may have been used in hundreds to thousands of patients (Mann et al. 2022).

6.10. Discovery Is Prioritized over Confirmation

New evidence was urgently needed in the COVID pandemic, and researchers were eager to present models based on small numbers and with inadequate validation of proposed models. The third update of the systematic review identified 208 new models and 24 external validations on July 7, 2020. The fourth and last update, focusing on prognostic models only, identified 381 new models and 225 external validations on February 17, 2021. The majority of developed models were never validated.

6.11. Education in Data Science Is Too Narrow

Publications on COVID-19 prediction models were generated under time pressure, such that many were not backed by teams that included sufficient methodological, statistical, and subject expertise. This likely contributed to the observation that only a handful of developed models were at low risk of bias.

6.12. Commercial Interests Challenge Transparency and Progress

Several companies who advertised their COVID-19 prediction models were contacted to obtain reports on model development and validation efforts, of which only one replied. The high-profile COVID-19 publications authored by this company were later retracted following allegations of having fabricated data (Piller & Travis 2020).

7. CONCLUSIONS

To some, clinical prediction modeling may appear to be a straightforward statistical task that requires little methodological training or planning, especially with statistical software being increasingly easy to use. The opposite is true: It is rare to achieve the ultimate goal of a successfully deployed prediction model with solid evidence of impact on patient outcomes. Achieving this goal is a tough challenge that has many enemies. It requires time, interdisciplinary expertise and collaboration, planning, and transparency. We have described a list of 12 enemies to successful prediction modeling that jeopardize its success. The enemies encompass scientific incentives within which researchers operate and inevitable challenges that underlie clinical data and study design, as well as poor methodology and reporting. Although large datasets are increasingly easy to obtain, it is much harder to obtain large datasets of sufficient quality and to use them properly for model development and evaluation. Ideally, an iterative process of validation and updating is followed as a

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learning healthcare system (Maddox et al. 2017), although researchers should simultaneously try to understand the reasons why the updates are necessary.

All of the challenges are actionable, but to varying degrees, as we illustrated with various positive examples. For example, there is no good excuse for not following reporting guidelines, but it is much harder to address challenges related to low signal-to-noise ratios and individual risk estimation. Apart from the individual researcher's responsibility, real improvement needs action from multiple stakeholders, including academic institutions, funding agencies, publishers, and regulators. Without such actions, we fear that much prediction research will remain a waste of time and money.

DISCLOSURE STATEMENT

B.V.C. has stock options in Gynaia, a company providing teaching, training, and decision support tools such as the ADNEX model for gynecologists and sonographers. The authors are not aware of any other affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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European Union AI Act: **https://digital-strategy.ec.europa.eu/en/policies/regulatory-framework-ai**

European Union Medical Device Regulations: **https://health.ec.europa.eu/medical-devices-new-regulations_en**

Evidencio: **https://evidencio.com/home**

PROBAST: **https://www.probast.org/**

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